Invasion, hybridization, and speciation: The breakdown of reproductive isolating barriers in the San Francisco Bay

Tim Legare[1] and Becky Fuller[1]
Department of Evolution, Ecology and Behavior, University of Illinois at Urbana-Champaign[1]


L. goodei


F1 hybrid


F1r hybrid


L. parva

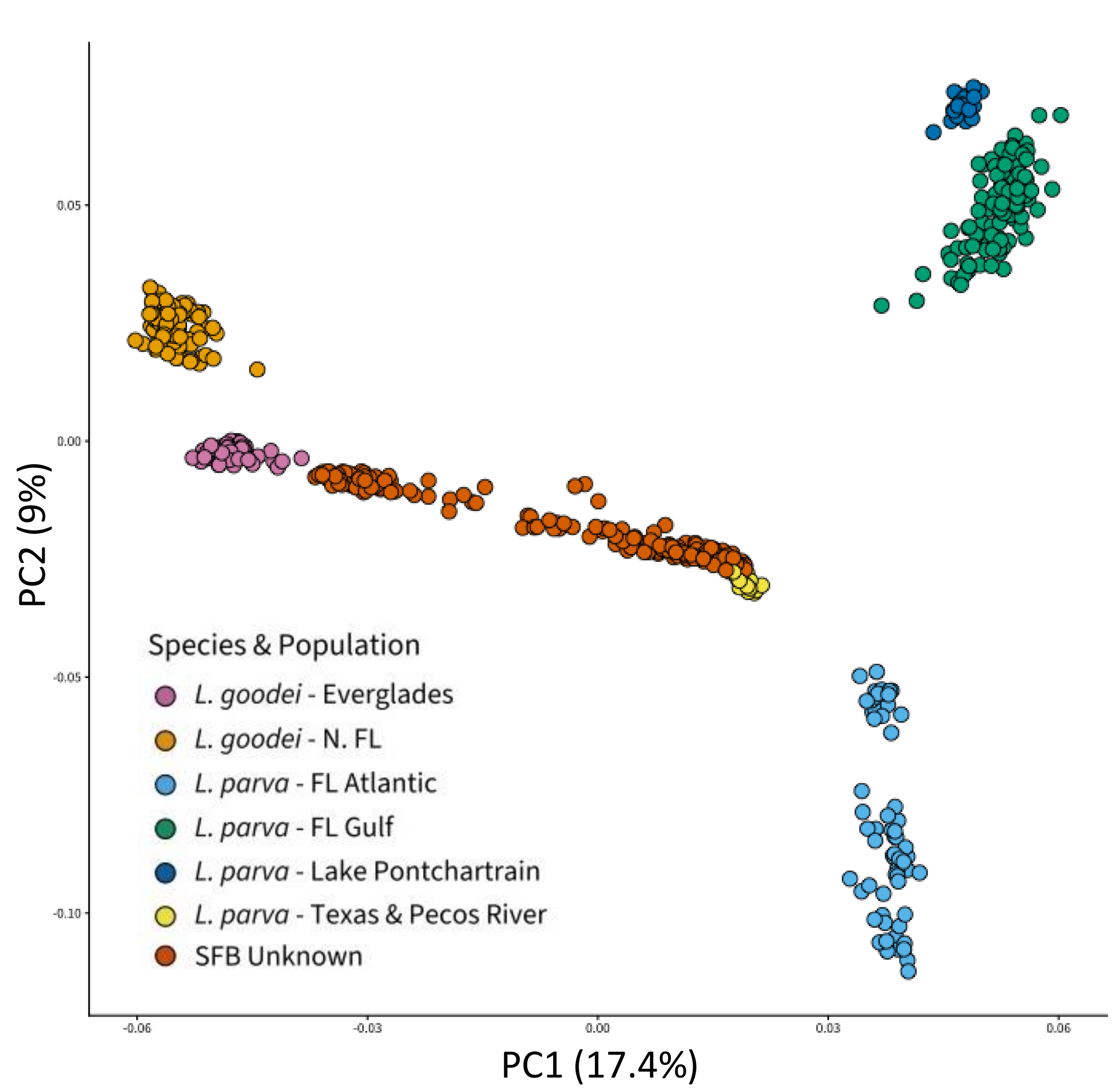
Introduction

- Hybridization between previously isolated lineages, once thought rare, is now recognized as a regular consequence of anthropogenic disturbance[1].
- Species introduction is a chief driver, rapidly eroding reproductive isolating barriers (RIBs) that took millions of years to evolve.
- Understanding which features of novel environments and invaders facilitate RIB breakdown is both a conservation concern and a window into how RIBs evolve, are maintained, and can be disrupted.
- *Lucania goodei* and *L. parva* are sister species with high behavioral isolation in native areas of sympatry, due to reinforcement[2].
- Both species have been introduced to the San Francisco Bay (SFB): *L. parva* in the 1950s (now widely abundant) and *L. goodei* in 2017[3]. Intermediate-appearing fish have recently been observed.
- **We ask: Has reproductive isolation broken down in the SFB, and if so, did the source fish originate from allopatric or sympatric populations?**

Methods

- Fish were collected from the SFB in 2024; Fish from FL, LA, and TX were collected in 2017.
- All were sequenced with RAD-Seq.
- Loci were called with STACKS. The STACKS populations module was run multiple times, followed by further filtering with vcftools and plink.
- PCA and ADMIXTURE were used to identify hybrids and determine populations of origin.

Results



Clear genetic differentiation between *L. goodei* and *L. parva*.

High levels of genetic differentiation among *L. parva* populations.

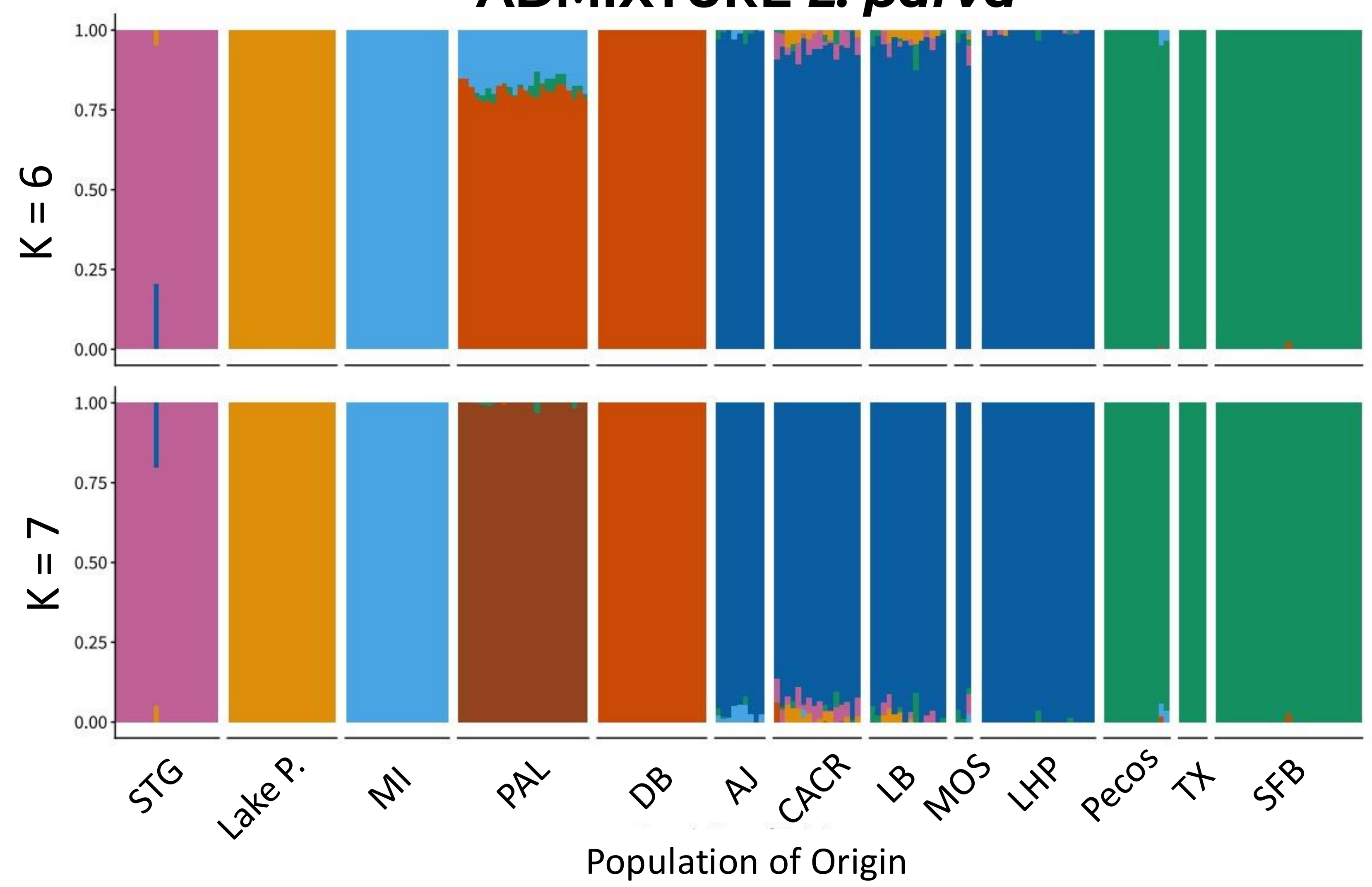
Hybrids are present in the SFB but are absent in the native range.

SFB *L. parva* cluster with *L. parva* from Texas/Pecos drainage.

SFB *L. goodei* appear similar to *L. goodei* from the Everglades but differences remain.

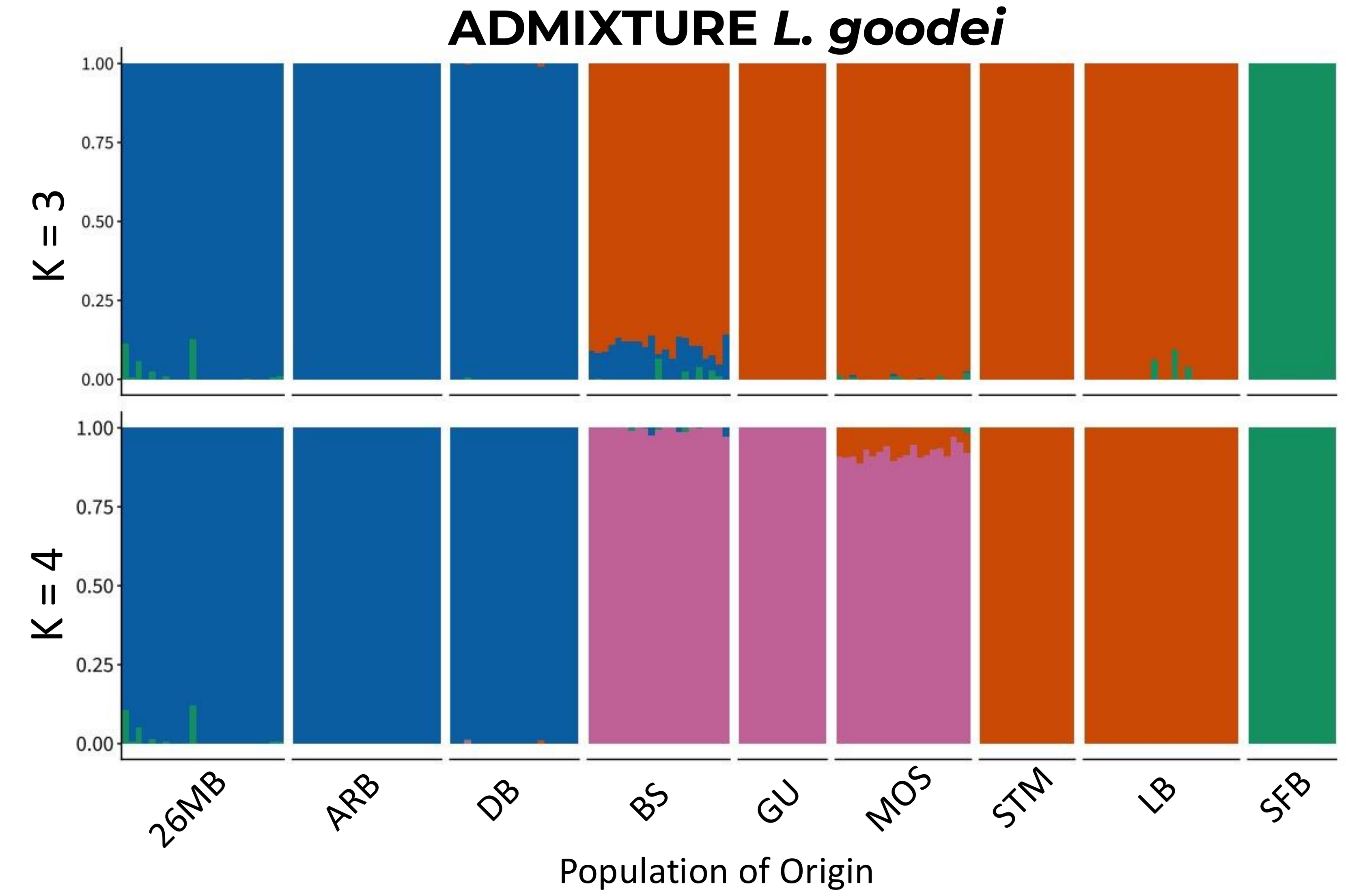
Results

ADMIXTURE *L. parva*



SFB *L. parva* clustered with Texas/Pecos River populations at the optimal K of 6 and this pattern was retained even after raising K to 7.

ADMIXTURE *L. goodei*



At the optimal K of 4, SFB *L. goodei* formed a cluster separate from all sampled Florida populations. This pattern persisted even upon reduction to a K of 3.

Results

Distribution of Identified Individuals

Discussion

•**Contemporary hybridization is prevalent and ongoing in the San Francisco Bay between *L. goodei* and *L. parva*.** There is little evidence for hybridization in the native range.

•The source population for *L. parva* is likely from the Pecos River Drainage, which is allopatric to *L. goodei*. The source for *L. goodei* is unclear but likely from South-West Florida.

•**Previous work shows that behavioral isolation is lowest when allopatric *L. parva* males interact with allopatric *L. goodei* females[2,4]. These conditions are at least partially met in this system.**

•The presence of admixed individuals throughout the study area suggests that the ecological conditions that disfavor hybrids in the native range are absent in the San Francisco Bay.

Future directions

Our next steps include determining **1)** which portions of the *L. goodei* and *L. parva* genomes have introgressed into back-crosses of each species, **2)** the extent to which cold, fresh water in the SFB favors introgression from *L. goodei* into *L. parva,* and **3)** which morphological traits best predict species and back-cross states.

Citations:
1. Rhymer, J. M. & Simberlorff (1996) *Annu. Rev. Ecol. Syst.* 27 **2.** Fuller, R. C., Mcghee, K. E. & Schrader (2007) *J. Evol. Biol.* 20 **3.** Mahardja, B. *et al.* (2020) *San Franc. Estuary Watershed Sci.* 18 **4.** St. John, M. E. & Fuller, R. C. (2021) *Curr. Zool.* 67